

Unit 4: Digital Forensics, Incident Response, and Cyber Law

Digital Forensics:

Digital Forensics is a branch of forensic science encompassing the recovery and investigation of material found in digital devices. It is governed by five main pillars:

1. **Identification:** Locating potential sources of evidence (devices, cloud storage, etc.).
2. **Preservation:** Ensuring data is not modified.
3. **Acquisition:** Creating a forensic duplicate of the data.
4. **Analysis:** Extracting and interpreting the data.
5. **Presentation:** Creating a report that is understandable to legal professionals.

2. Taxonomy of Digital Evidence

Evidence is categorized based on its stability and where it resides within the hardware architecture.

A. Volatile Data (The "Live" State)

This data exists only as long as the device is powered. In a university exam, you should mention the **Order of Volatility (OOV)**, which dictates that you collect this first.

- **Registers & Cache:** The fastest, most volatile data.
- **Routing Tables/ARP Cache:** Information about network traffic and connected devices.
- **Process Table:** Currently running applications (including hidden malware).
- **Kernel Statistics:** System-level performance and state data.
- **Main Memory (RAM):** Contains passwords, decrypted files, and clipboard content.

B. Non-Volatile Data (The "Persistent" State)

Data that survives a power cycle. This is usually the bulk of the investigation.

- **Internal/External Storage:** HDD, SSD, SD cards, and USB drives.
- **Firmware:** Data stored in ROM or flash memory (e.g., BIOS/UEFI settings).
- **Hidden Partitions:** Areas of a disk not mapped by the operating system.

C. Specialized Evidence Forms

- **Metadata:** Often called "the smoking gun." It includes **MAC Times** (Modified, Accessed, Created) which help build a timeline.
- **Unallocated Space:** Areas of the disk where files were "deleted." The data remains until overwritten, allowing for **File Carving**.

3. Evidence Acquisition Methods

Acquisition is the "imaging" phase. It must be performed using a **Write Blocker** to prevent any accidental changes to the original media.

A. Live vs. Static Acquisition

Feature	Live Acquisition	Static (Dead) Acquisition
System State	System is powered ON.	System is powered OFF.
Data Captured	Captures RAM, encrypted volumes, and active network connections.	Captures the physical disk bit-by-bit.
Risk Level	High (malware may delete data if it detects an investigator).	Low (the data state is frozen).
Use Case	Cloud servers, encrypted laptops, or running malware.	Standard workstation or mobile device forensics.

B. Physical vs. Logical Acquisition

- **Physical Acquisition:** A bit-stream image (exact clone) of the entire physical media. This includes the boot sector, unallocated space, and slack space. It allows for the recovery of deleted files.
- **Logical Acquisition:** Only copies specific files or directories (e.g., just the "Documents" folder). This is faster but misses deleted data and hidden system files.
- **Sparse/Targeted Acquisition:** A hybrid method used when time is limited, collecting only specific file types (e.g., all .jpg and .pdf files) across the disk.

2. Evidence Preservation and Chain of Custody

A. Evidence Preservation: Maintaining Integrity

Preservation is the process of protecting digital evidence from any alteration, damage, or degradation from the moment it is identified until it is presented in court.

Key Techniques and Technologies:

- **Write Blockers (Hardware and Software):** These are essential tools that sit between the evidence (source) and the forensic workstation (destination). They allow "read" commands but physically or logically intercept and block "write" commands, ensuring the original device remains untouched.
- **Forensic Imaging (Bit-Stream Copy):** Investigators never work on the original device. They create a "Forensic Image"—a bit-for-bit clone that includes hidden files and unallocated space.
- **Cryptographic Hashing:** This is the "digital seal." Algorithms like **SHA-256** or **MD5** generate a unique mathematical value for the evidence.
 - **Verification:** If the hash of the image matches the hash of the original source, it proves the data has not changed by even a single bit.
- **Physical Protection:** Storing devices in **Faraday Bags** (to block remote wiping signals for mobile devices) and climate-controlled, anti-static environments to prevent hardware failure.

B. Chain of Custody (CoC): The Paper Trail

The **Chain of Custody** is a chronological, written record that accounts for the seizure, custody, control, transfer, analysis, and disposition of physical and electronic evidence.

1. Essential Elements of a CoC Document

To be legally defensible, a CoC form must contain:

- **Unique Case Identifiers:** Case number and evidence tag numbers.
- **Description of Evidence:** Make, model, serial number, and physical condition of the device (e.g., "iPhone 14 with cracked screen").
- **The "Who, When, and Where":** Every person who touched the evidence must sign the log with a specific timestamp and the reason for handling (e.g., "Transferred from Scene to Lab for Analysis").

- **Storage Locations:** Records of the secure locker or safe where the evidence is kept when not being analyzed.

2. Critical Importance of CoC

- **Admissibility in Court:** Under legal standards (like the Daubert Standard or Frye Standard), evidence is only admissible if the prosecution can prove it was not tampered with. A "break" in the chain (missing signatures or unaccounted time) often leads to evidence being suppressed.
- **Accountability:** It prevents the "planting" of evidence and ensures that the investigator is responsible for the items in their possession.
- **Quality Assurance:** It allows independent third parties to audit the investigation and verify that standard operating procedures were followed

3. File System and Log Analysis

A. File System Analysis

A file system is the method and data structure that an operating system uses to keep track of files on a disk. Forensics focuses on recovering data that the OS "thinks" is gone.

1. Common File Systems:

- **NTFS (New Technology File System):** Used by Windows. Its heart is the **Master File Table (MFT)**, which contains records for every file and directory.
- **EXT4 (Fourth Extended Filesystem):** The standard for Linux. It uses **Inodes** to store metadata about files.
- **APFS (Apple File System):** Optimized for SSDs/Flash storage used in macOS and iOS.
- **FAT32/exFAT:** Common in USB drives and SD cards due to universal compatibility.

2. Key Forensic Concepts:

- **Master File Table (MFT):** In Windows forensics, the **MFT is the most important artifact. Even if a file is deleted, its MFT record may remain, showing when it was created and where it sat on the disk.**
- **Metadata Analysis:** Examining the **"hidden" details such as GPS coordinates in a photo (EXIF data) or the "Last Printed" date in a Word document.**
- **File Carving:** This is the **process of recovering files based on their File Headers** (Magic Bytes). For example, a JPEG file always starts with the hex values FF D8 FF. Even if the file system is formatted, a tool can "carve" out the image by looking for these signatures.

- **Slack Space:** Files are stored in fixed-size "clusters." If a file is smaller than the cluster, the remaining space is "slack." **Suspects often hide small fragments of data or passwords in this unused space because the OS doesn't show it to the user.**

B. Log Analysis

Logs are chronological records of events generated by the OS, applications, or network devices. They are the primary tool for **reconstructing a timeline**.

1. Types of Logs and Their Forensic Value:

- **System Logs:** Record hardware changes, device drivers, and system startup/shutdown times. *Value: Proving if a USB drive was plugged in at a specific time.*
- **Security/Audit Logs:** Record login attempts (successful and failed), privilege escalations, and resource access. *Value: Identifying brute-force attacks or unauthorized access.*
- **Application Logs:** Generated by software like web browsers or databases. *Value: Showing what a user searched for or what data was modified in a database.*
- **Network Logs:** Records of connections (IP addresses, ports, and protocols). *Value: Proving communication with a malicious Command & Control (C2) server.*

2. The Goal: Timeline Reconstruction

The ultimate goal of log analysis is to answer the **5 W's**:

1. **Who** logged in?
2. **What** files did they access?
3. **Where** did they connect from?
4. **When** did the activity occur?
5. **Why/How** did the system crash or leak data?

C. Challenges in Analysis

For extra marks, mention the hurdles an investigator faces:

- **Log Tampering:** Advanced attackers often delete logs to hide their tracks. Forensics involves looking for "gaps" in the log sequence.
- **Clock Skew:** If the system clock was wrong, the logs will show the wrong time. Investigators must calculate the "drift" to sync events correctly.

- **Volume:** Modern systems generate millions of log entries; tools like **SIEM** (Security Information and Event Management) are used to filter the "noise."

4. Forensic Imaging and Tools

A. Forensic Imaging: The "Gold Standard" of Copying

Forensic imaging is the process of creating a **bit-stream image** of a storage medium. Unlike a standard "copy-paste" or "backup," forensic imaging captures every single bit of data on the drive.

1. Types of Images:

- **Physical Image:** An exact duplicate of the entire physical drive, from the first sector to the last. This includes the boot sector, partition tables, and **unallocated space** (where deleted files live).
- **Logical Image:** Captures only the files and folders visible to the operating system. This is used when a full physical image is impossible (e.g., due to time constraints or legal warrants restricted to specific folders).
- **Targeted (Sparse) Image:** Collects specific file types (e.g., only .docx and .pdf files) or data from a specific timeframe.

2. Crucial Imaging Concepts:

- **Bit-Stream Copy:** A sector-by-sector copy that ensures hidden data (like slack space) is captured.
- **Hash Verification:** Immediately after the image is created, a hash (MD5/SHA-1/SHA-256) is calculated. If the image's hash matches the original drive's hash, the image is considered **forensically sound**.
- **Compression:** Forensic images can be compressed (using formats like .E01) to save space without losing data integrity.

B. The Digital Forensics Toolkit

In an exam, you should categorize tools by their function. Mentioning both open-source and commercial tools demonstrates a well-rounded knowledge.

1. Disk Imaging & Preview Tools

- **FTK Imager (AccessData):** A free, industry-standard tool for creating forensic images and viewing the contents of a drive without changing the metadata.

- **Guymager:** A popular open-source, Linux-based imaging tool often found in forensics-focused OS distributions.

2. Comprehensive Analysis Suites

- **Autopsy / The Sleuth Kit:** The leading **open-source** platform. It allows for timeline analysis, keyword searching, and web artifact extraction.
- **EnCase:** A powerful commercial suite used globally by law enforcement. It is known for its advanced reporting and "Evidence File" (.E01) format.
- **X-Ways Forensics:** Known for being highly efficient, fast, and portable (it can run from a USB stick).

3. Specialized Tools

- **Volatility:** The primary tool for **Memory (RAM) Forensics**. It allows investigators to see what was happening on the computer at the exact moment of seizure.
- **Wireshark:** A network protocol analyzer used to capture and interactively browse the traffic running on a computer network.
- **Cellebrite / UFED:** The gold standard for **Mobile Forensics** (extracting data from smartphones and tablets).

C. Comparison Table: Commercial vs. Open Source

Feature	Commercial (EnCase/FTK)	Open Source (Autopsy)
Cost	Expensive (License-based)	Free
Support	Professional technical support	Community-driven (forums/wikis)
Validation	Heavily vetted for court use	Widely accepted, but requires manual validation
Features	Often "all-in-one" automated suites	Modular; may require multiple tools

Email and network traffic forensics:-

Email Forensics: Investigating Digital Correspondence

Email forensics involves analyzing the source, content, and transmission path of an email to verify its authenticity and trace its origin.

1. The Email Architecture

An investigator must understand how an email travels to identify where logs are stored:

- **MUA (Mail User Agent):** The client-side application (e.g., Outlook, Apple Mail).
- **MTA (Mail Transfer Agent):** The server that routes the mail (e.g., Postfix, Microsoft Exchange).
- **MDA (Mail Delivery Agent):** Delivers the mail to the user's specific mailbox.

2. The Anatomy of an Email Header

The header is the most vital piece of evidence. It is analyzed **bottom-to-top**, as the bottom-most "Received" entry is generally the first server to handle the message.

- **Received Fields:** Shows the IP addresses of every server the email passed through.
- **Message-ID:** A unique string assigned by the mail system.
- **X-Headers:** Non-standard headers that can reveal the specific software used by the sender.

3. Verification & Authentication Protocols

In cases of spoofing or phishing, three DNS-based records are examined:

- **SPF (Sender Policy Framework):** Validates that the sending IP is authorized by the domain owner.
- **DKIM (DomainKeys Identified Mail):** Uses a cryptographic signature to ensure the email content was not altered.
- **DMARC:** A policy framework that instructs the receiver how to handle emails that fail SPF or DKIM.

Network Traffic Forensics: Analyzing Data in Motion

Network forensics deals with volatile data that is transmitted across a network. It is often used for detecting data exfiltration, unauthorized access, and malware "beaconing."

1. Evidence Acquisition Methods

- **Full Packet Capture (FPC):** Capturing the entire payload and header (PCAP files). This allows for complete **Session Reconstruction** (e.g., rebuilding a downloaded file).
- **NetFlow Analysis:** Captures only the "summary" (Source/Dest IP, Ports, Timestamps). It is used for high-volume networks where storing full packets is impossible.

2. Key Forensic Techniques

- **Packet Sniffing:** Real-time monitoring of traffic using a NIC in "Promiscuous Mode."
- **Deep Packet Inspection (DPI):** Examining the data part (and header) of a packet as it passes an inspection point to search for non-compliance, viruses, or spam.
- **Protocol Analysis:** Identifying anomalies, such as sensitive data being sent over an unencrypted protocol like HTTP or Telnet.

Tools of the Trade

A high-scoring exam answer should categorize tools by their specific utility:

Email Forensic Tools:

- **Forensic Email Collector (FEC):** Specifically designed to preserve emails from cloud providers (Gmail, O365) while maintaining metadata.
- **MailXaminer:** An advanced tool for searching and analyzing large volumes of email archives.
- **MxToolbox:** A web-based utility used to verify SPF/DKIM records and blacklist status.

Network Forensic Tools:

- **Wireshark:** The industry-standard GUI for analyzing PCAP files and protocol dissection.
- **tcpdump:** A powerful command-line packet analyzer for Linux/Unix systems.
- **NetworkMiner:** A passive network sniffer that can automatically extract files (images, documents) from a network capture.
- **Zeek (Bro):** A network security monitor that translates raw traffic into high-level event logs.

Challenges and Legal Considerations

Forensic investigators face several hurdles that must be addressed in a professional report:

1. **Encryption (TLS/SSL):** The widespread use of HTTPS hides the payload. Investigators must rely on **SSL Inspection** or analyze traffic patterns/metadata.
2. **Log Tampering:** Attackers often delete local logs. This highlights the importance of **Centralized Logging (SIEM)** for evidence preservation.
3. **Anti-Forensics:** Techniques like IP spoofing, VPNs, and Tor make tracing the true origin of a network attack significantly more difficult.
4. **Admissibility:** For network evidence to hold up in court, the investigator must prove the **Chain of Custody** and ensure the capture tool did not inadvertently "drop" packets, leading to an incomplete record.

Comparison :-

Feature	Email Forensics	Network Forensics
Primary Goal	Authentication & Source Tracing	Incident Response & Traffic Analysis
Key Artifact	.eml, .pst files, MIME Headers	.pcap files, Flow logs, Firewall logs
Best Practice	Analyze headers bottom-to-top	Maintain a continuous capture during an event
Typical Use Case	Phishing, Harassment, Fraud	Intrusion Detection, Data Leaks

Incident Response

1. Incident Detection and Reporting

The first phase of the incident response life cycle focuses on identifying a potential security breach and notifying the correct stakeholders.

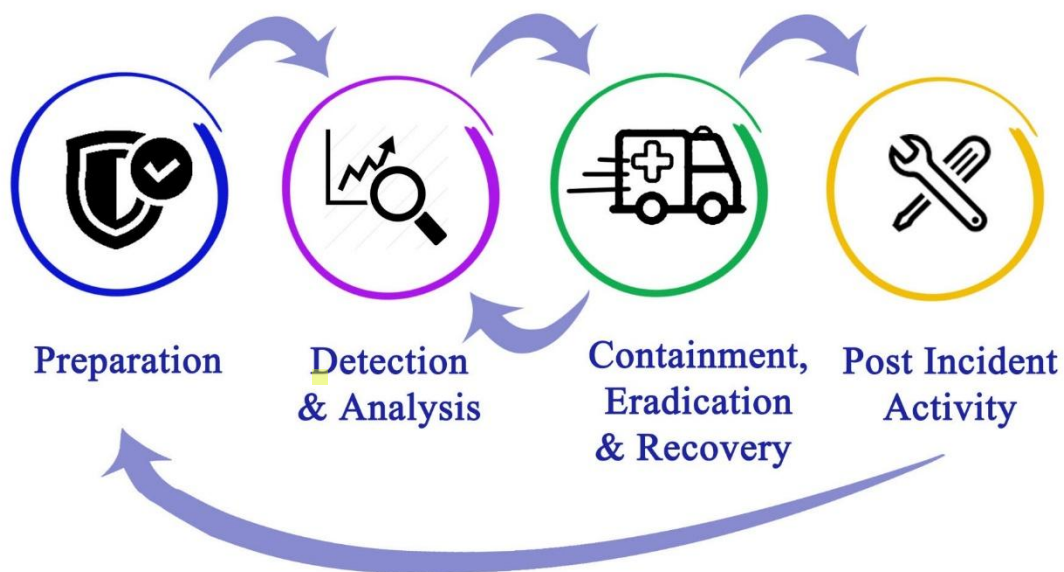
- Detection Sources:
 - **Host-based:** Monitoring system logs, file integrity checkers, and antivirus alerts.
 - **Network-based:** Analyzing traffic via IDS (Intrusion Detection Systems), SIEM (Security Information and Event Management) platforms, and firewall logs.

- **User Reporting:** Establishing a "Whistleblower" or helpdesk channel where employees report anomalies (e.g., suspicious emails or slow system performance).
- Reporting Protocol:
 - Initial Triage: Categorizing the incident by severity (Low, Medium, High, Critical).
 - Escalation Matrix: Determining who needs to be informed (e.g., IT Manager, CISO, Legal Team, or Law Enforcement) based on the nature of the data compromised.

2. Incident Handling Process

Once an incident is confirmed, the organization follows a structured containment, eradication, and recovery (CER) workflow.

Incident Response Planning



A. Containment

The primary goal is to "stop the bleeding" and prevent the threat from spreading to other systems.

- **Short-term Containment:** Immediate actions like isolating a compromised workstation from the network or disabling a compromised user account.
- **Long-term Containment:** Applying temporary fixes (like blocking specific IP addresses at the firewall) to allow systems to continue operating while a permanent solution is developed.

B. Eradication

After the threat is contained, the root cause must be removed.

- **Malware Removal:** Running specialized cleaners or deleting malicious scripts.
- **Vulnerability Patching:** Identifying how the attacker got in and closing that "hole" (e.g., patching a known software bug or changing weak passwords).
- **System Re-imaging:** In many cases, it is safer to wipe a drive completely and reinstall the OS from a known-clean image.

C. Recovery

Restoring systems to normal production operation.

- **Restoration:** Bringing systems back online, often starting with the least-critical systems first.
- **Validation:** Testing and monitoring the restored systems to ensure they are functioning correctly and are no longer compromised.

3. Post-Incident Analysis and Lessons Learned

Often called a "Post-Mortem," this is arguably the most important phase for long-term security.

- **Documentation:** Creating a detailed timeline of the "Who, What, Where, When, and How" of the incident.
- **Gap Analysis:** Identifying where the security controls failed. Was it a technical failure or a human error?
- **Strategy Update:** Modifying the Incident Response Plan (IRP) and security policies to prevent a recurrence. This might involve additional student/staff training or investing in new defensive tools.

4. Forensic Auditing Techniques

Forensic Auditing is the application of investigative and accounting proficiency to extract digital evidence that is suitable for use in a court of law. It combines **Digital Forensics** (the technical process of collecting and preserving evidence) with legal protocols to ensure the integrity of an investigation.

Core Procedural Principles (The "Audit" Side)

Before technical analysis begins, strict legal standards must be met to ensure the evidence is admissible:

- **Data Acquisition (Write-Blocking):** Investigators use hardware or software **write-blockers** to ensure the original evidence is never modified. A "bit-stream" image (an exact 1:1 copy) is created for analysis.
- **Chain of Custody:** A chronological, written log must be maintained. it tracks every person who handled the evidence, the time they had it, and the purpose of their access. If the chain is broken, the evidence is legally void.

Key Forensic Techniques (The "Technical" Side)

Technical investigation is categorized based on where the data is stored:

Technique	Scope of Investigation	Practical Example
Disk Forensics	Recovery of data from hard drives or SSDs using File Carving to find fragments in unallocated space.	Recovering "permanently deleted" emails or documents.
Memory (RAM) Forensics	Analyzing volatile data that disappears when a computer is turned off.	Detecting hidden malware or encryption keys currently in use.
Network Forensics	Monitoring and analyzing network traffic and packets to identify intruder behavior.	Finding the source IP address of an unauthorized login attempt.
Log Correlation	Matching timestamps across web servers, databases, and firewalls to reconstruct a timeline.	Reconstructing the exact path an attacker took through a network.

Advanced Analysis Methods

- **Cross-Drive Analysis:** Correlating information found on multiple different devices (e.g., a laptop and a mobile phone) to find matching patterns or communication.
- **File Integrity Checking:** Using cryptographic hashes (like SHA-256) to detect if a system file has been tampered with or replaced by a Trojan horse.

A successful forensic audit is not just about technical skill; it is about the **integration of technology and law**. While techniques like disk and memory forensics find the "smoking gun," procedures like the Chain of Custody ensure that the "smoking gun" can be used to convict a criminal in a legal proceeding.

Cyber Law and Ethics

1. Computer Crime Laws and Regulations

Definition: Cybercrime refers to any illegal act where a computer or network is the tool, the target, or the scene of the crime. Cyber Law is the legal infrastructure that provides justice to victims and punishment for digital offenders.

Classification of Cybercrime

1. Cyber-Dependent Crimes: Crimes that **can only be committed using computers and networks** (e.g., Hacking, DDoS attacks, Malware distribution).
2. Cyber-Enabled Crimes: Traditional crimes "**scaled up**" by the internet (e.g., Online Fraud, Identity Theft, Cyberstalking).

Key Legal Frameworks

- India: **Information Technology (IT) Act, 2000**. Key sections include Section 43 (Damage to computer systems) and Section 66 (Computer-related offenses/hacking).
- USA: **Computer Fraud and Abuse Act (CFAA)**, which focuses on unauthorized access to protected computers.

2. Data Protection Laws (GDPR & DPDP)

Definition: Regulatory frameworks that dictate how "Personal Data" (Name, Biometrics, IP Addresses) is handled to ensure individual privacy.

The GDPR (General Data Protection Regulation)

The world's strictest privacy law, centered on 7 Core Principles:

- **Lawfulness, Fairness, & Transparency**: Clear legal basis for processing.
- **Purpose Limitation**: Use data only for the stated reason.
- **Data Minimization**: Collect only the "need-to-know" data.
- **Storage Limitation**: Do not keep data longer than necessary.

- **Integrity & Confidentiality**: Ensure high-level security.

Indian Context: DPDP Act (2023)

The Digital Personal Data Protection Act is India's modern framework. It emphasizes Consent, Data Fiduciaries (those who store data), and Significant Data Processors (large tech companies).

3. Intellectual Property (IP) in Cyberspace

IP laws protect intangible digital assets from theft and unauthorized replication.

Type	Protects	Example in Cyber
Copyright	Original expressions/content	Software source code, Website UI, Videos.
Trademark	Brand identity	Domain names, Logos. (Avoids <i>Cybersquatting</i>).
Patent	Technical inventions	Encryption algorithms, novel hardware designs.

4. Digital Evidence in Court

Definition: **Any information of probative value that is stored or transmitted in binary form.**

The "Gold Standard" for Admissibility

For evidence to be accepted in court, it must meet four criteria:

1. **Authenticity**: Evidence must be tied to a specific author or system.
2. **Integrity**: Must prove the file was not tampered with (verified via Hash Values like SHA-256).
3. **Chain of Custody**: A document showing the chronological history of who handled the evidence.
4. **Reliability**: The computer system used to extract the data must be functioning correctly.

5. Ethical Hacking vs. Malicious Hacking

While the technical skills used are identical, the intent and legality create a binary distinction.

- Ethical Hacking (White Hat): Performed with **written permission** (Rules of Engagement) **to identify vulnerabilities and strengthen security.**

- Malicious Hacking (Black Hat): Unauthorized access for personal gain, corporate espionage, or system disruption.

Feature	Ethical Hacking (White Hat)	Malicious Hacking (Black Hat)
Intent	To improve security and find flaws.	To steal, damage, or disrupt.
Permission	Explicit written consent (Rules of Engagement).	None (Illegal).
Method	Follows legal & ethical frameworks.	Uses exploits, social engineering, and stealth.
Outcome	Detailed report and vulnerability patch.	Data breach, financial loss, or system crash.

6. Responsible Disclosure

This is the ethical middle ground between discovering a bug and keeping it secret.

The Standard Workflow

1. **Discovery**: Researcher finds a vulnerability.
2. **Private Reporting**: Researcher alerts the vendor privately.
3. **Grace Period**: A window (usually 90 days) for the vendor to develop a patch.
4. **Patch Deployment**: The fix is released to the public.
5. **Public Disclosure**: The researcher shares the technical details for community learning.

Cyber Crime:-

I. Fundamentals of Cybercrime & Cyber Terrorism

Definition: Cybercrime is any criminal activity that involves a computer, networked device, or a network. Cyber Law is the legal framework that protects users and regulates digital transactions.

1. Classification of Cybercrime

Category	Focus	Primary Examples
Against Individuals	Personal safety & privacy	Phishing, Cyberstalking, Identity Theft.
Against Property	Digital assets/finance	Ransomware, Credit Card Fraud, IPR Theft.
Against Organizations	Business continuity	DDoS Attacks, Corporate Espionage, Hacking.
Against Society/Gov	National stability	Cyber Terrorism, Spreading Fake News/Panic.

2. Cyber Terrorism

- Definition:** The use of computer network tools to shut down critical national infrastructure (energy, transportation, government operations) to coerce or intimidate a government or civilian population.
- Key Distinguishing Factor: Motive.** Unlike general cybercrime (profit/revenge), terrorism is driven by political, religious, or ideological goals.

Comparison between Cybercrime and Cyber Terrorism

Feature	Cybercrime	Cyber Terrorism
Primary Goal	Profit or Personal Gain	Fear, Chaos, or Political Change
Primary Target	Individuals or Businesses	Critical National Infrastructure
Scale of Impact	Usually Localized	National or Global
Example	Phishing for Bank Details	Shutting down a National Power Grid

II. Motives and Impacts

Motives (The "Drives")

- **Financial Gain:** The most common motive (Ransomware, Fraud).
- **Political/Ideological:** Promoting a cause or disrupting an enemy state (Hacktivism).
- **Revenge:** Insider threats (disgruntled employees).
- **Recognition:** "Script kiddies" or hackers looking for fame in the community.

Impacts (The "Consequences")

Type of Impact	Description
Economic	Cost of recovery, loss of revenue, and legal fines.
Reputational	Loss of customer trust and brand value.
Operational	System downtime and disruption of essential services.
Psychological	Trauma for victims of stalking or identity theft.

III. Legal Framework: India & Global

1. The IT Act 2000 (Amended 2008)

- **Section 43:** Penalty for damage to computer systems without permission.
- **Section 66:** Punishment for hacking and computer-related offenses.
- **Section 66F:** Specifically deals with **Cyber Terrorism** (Life imprisonment).

2. Data Protection: GDPR vs. DPDP (2023/2026)

- **GDPR (Global):** Focuses on 7 principles (Lawfulness, Minimization, Purpose Limitation, etc.) and user rights (Right to be Forgotten).
- **DPDP Act (India):** India's modern privacy law.
 - **Key Duty:** Organizations must notify the **Data Protection Board** and users of a breach **"without delay"** (typically interpreted as within **72 hours**).
 - **Penalty:** Can reach up to **₹250 Crore** for major violations.

IV. Intellectual Property Rights (IPR) in Cyberspace

Type	Protects	Cyber Example	Duration
Copyright	Creative Expression	Software Code, Videos, Blog Posts	Life + 60-70 Years
Trademark	Brand Identity	Domain Names, App Logos	Indefinite (Renewable)
Patent	Technical Invention	Encryption Algorithms, Hardware	20 Years

Concept to Note: **Cybersquatting** is the illegal practice of registering a trademarked domain name (e.g., www.google-support.com) to sell it later or mislead users.

Reporting and Countermeasures

This is the "proactive" part of the notes.

A. Reporting Mechanism (The Protocol)

1. **Internal Reporting:** Notify the organization's Incident Response Team (IRT).
2. **External Reporting:** * **India:** [suspicious link removed] or local Cyber Cells.
 - o **Global:** Specialized agencies like CERT (Computer Emergency Response Team).
3. **Evidence Preservation:** Do not shut down the system immediately; take screenshots, save logs, and maintain the **Chain of Custody**.

B. Countermeasures (Defense-in-Depth)

- **Technical:** * **Firewalls & IDS/IPS:** To monitor and block malicious traffic.
 - o **Encryption:** Protecting data at rest and in transit.
 - o **Multi-Factor Authentication (MFA):** Adding layers beyond passwords.
- **Administrative:** * **Security Audits:** Regular vulnerability assessments.
 - o **Employee Training:** Reducing "Human Error" (the weakest link).
- **Legal:** Enforcement of IT Acts and international treaties.